Department of Computer Science and Engineering
University College of Engineering (A), Osmania University

B.E (CSE) V-Semester II-Internal Examination-2024

Subject: NUMBER THEORY AND CRYPTOGRAPHY

Time: 1:30 Hr. **Max. Marks: 30**

Date: 24-12-2024

Note: Answer all Question in Part-A and any two in Part-B

Q.NO		Marks	BT	CO
	Part –A (5 x 2 = 10 Marks)			
1	List various requirements of authentication.	2	1	2
2	What is Digital Certificate?	2	2	2
3	What are the features of SET?	2	2	3
4	What is Proxy Firewall?	2	1	2
5	For SSH packets, what is the advantage, if any, of not including the MAC in the scope of the packet encryption?	2	2	3
	Part –B (2*10 = 20 Marks)			
4 (a)	Explain how a signature is signed and verified in Digital Signature Algorithm (DSA)?	5	4	4
(b)	In detail explain the process involved in generating a 160 bit message digest using SHA-1 algorithm.	5	4	3
5 (a)	Draw the Pretty Good Privacy (PGP) Message Generation and Message Reception Architecture. Explain.	5	3	3
(b)	What are the core components of a PKI? Briefly describe each component.	5	5	2
6 (a)	Describe Transport tunnel modes used for IPsec AH authentication bringing out their scope relevant to IPV4.	5	1	3
(b)	What is SSL record Protocol? What are the steps involved in SSL record Protocol.	5	3	2